Introduction and Usage of the gem5 Simulator

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What is gem5?

Definition: gem5 is a modular, open-source computer architecture simulation platform supporting diverse CPU, cache, memory, and system configurations.

- Multi-ISA support: x86, ARM, RISC-V, MIPS, etc.
- Two simulation modes: Full System (FS) and Syscall Emulation (SE)
- Modular architecture: CPU / Cache / Memory / Devices are composable
- Hybrid design: Python for configuration, C++ for simulation core

Major Subsystems Overview

gem5 is organized into several major subsystems that can be independently configured and extended.

Subsystem	Purpose	Key Components
CPU Models	Processor simulation	SimpleCPU, O3CPU, MinorCPU
Memory System	Memory hierarchy	Cache models, Ruby coherence, DRAM controllers
ISA Support	Instruction sets	ARM, x86, RISC-V decoders and execution
Device Models	I/O and platform	Platform models, device controllers
System Modes	Execution context	Full-system, syscall emulation

Composition via SimObjects

Each subsystem is implemented through the SimObject framework, allowing flexible composition of different models. For example, an ARM system can use the O3 CPU model with Ruby coherence and DDR4 memory controllers.

WebUI Usage

Step 1: Open GitHub Codespaces

- Launch a Codespace for this repository.(5-15 minutes)

Step 2: Start the Web UI

```
python3 configs/class/webui.py --host 127.0.0.1 --port 8080
```

Step 3: Configuration Selection

- Select a predefined configuration
- or build a custom configuration

Step 4: Analyze results

- Inspect statistics (IPC, miss rates), logs, and outputs

Course Experiments and Test Programs

Experiments (`configs/class/experiments/`)

1. capacity
2. associativity
3. line size
4. replacement policy
5. hierarchy
6. is_read_only
7. writeback_clean

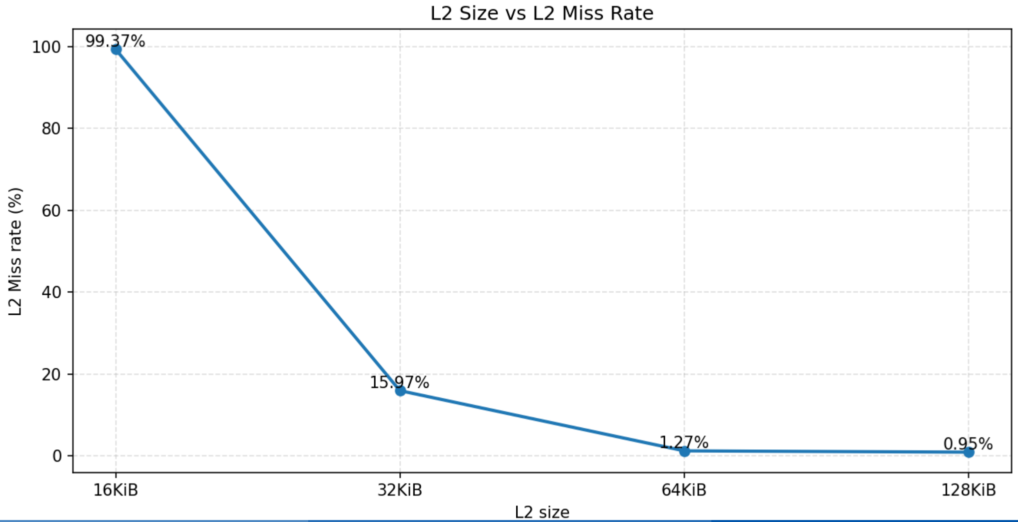
Test programs (`tests/class/bin/x86/`)

- hello, matrixmul, matrixmul_smallsize
- memsum, streamcopy, cachethrash
- fpvector, branchstorm

Live Demo

https://github.com/wilburx813/gem5

Results Demo:L2 Capacity vs L2 Miss Rate



Summary and References

Summary

- gem5 is a powerful, extensible tool for architecture education and research
- Modular, multi-ISA, configurable from simple to system-level studies

References

- gem5 website: https://www.gem5.org
- Documentation: https://www.gem5.org/documentation/
- GitHub: https://github.com/gem5/gem5
- Recommended: *Learning gem5* official tutorials

Q&A

Questions?